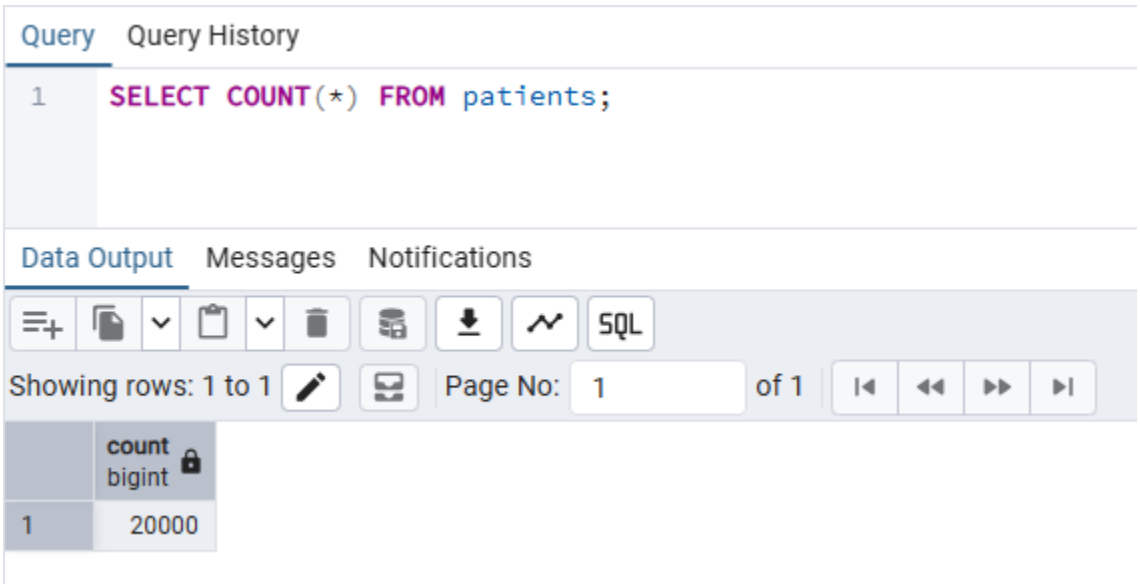
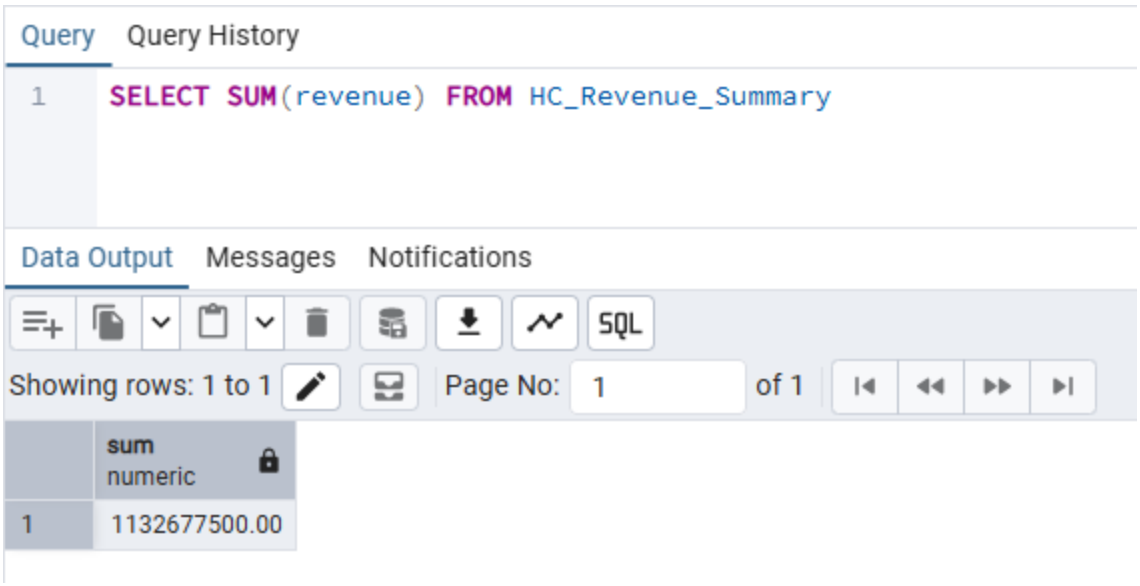
Analytical Testing, KPI Validation & Dashboard Refinement

1. KPI Validation & Database Reconciliation
To ensure institutional-grade reliability, core executive and operational metrics were audited and cross-reconciled against raw PostgreSQL staging tables (patients, HC_Revenue_Summary, and HC_Patient_Feedback):

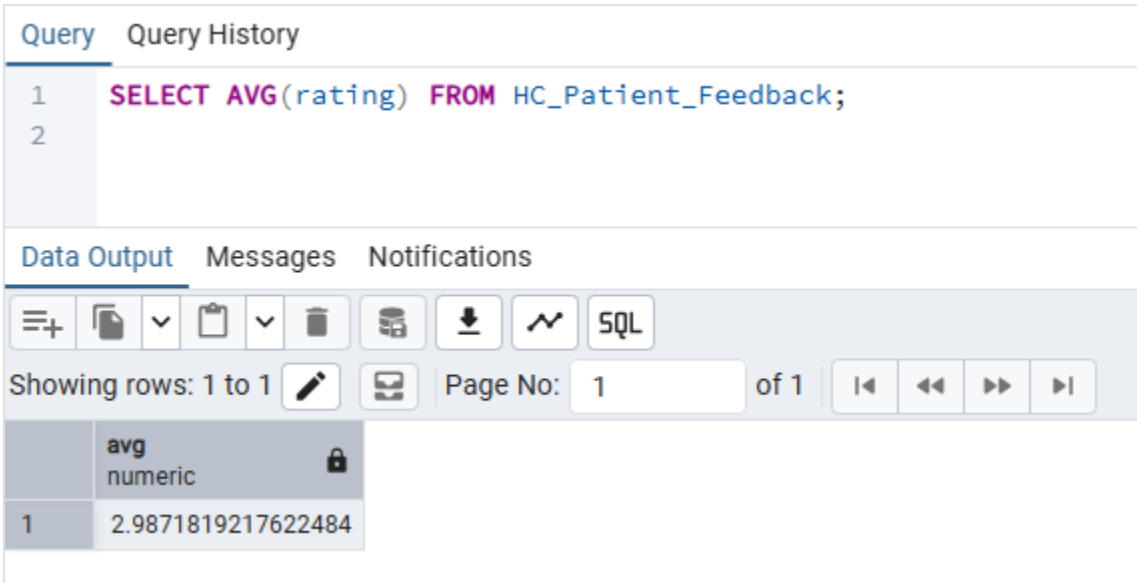
- **Total Registered Patients (**20K**):** Validated via SQL query SELECT COUNT(*) FROM patients;, which returned an exact count of 20000. This confirms complete alignment between the backend patient registry and the **Patient Analytics** dashboard.



- **Total Revenue (**3.1bn**):** Audited via SQL query SELECT SUM(revenue) FROM HC_Revenue_Summary;, returning 1,132,677,500.00. This verifies mathematical consistency across the macro-level financial aggregates displayed on the **Financial Dashboard**.



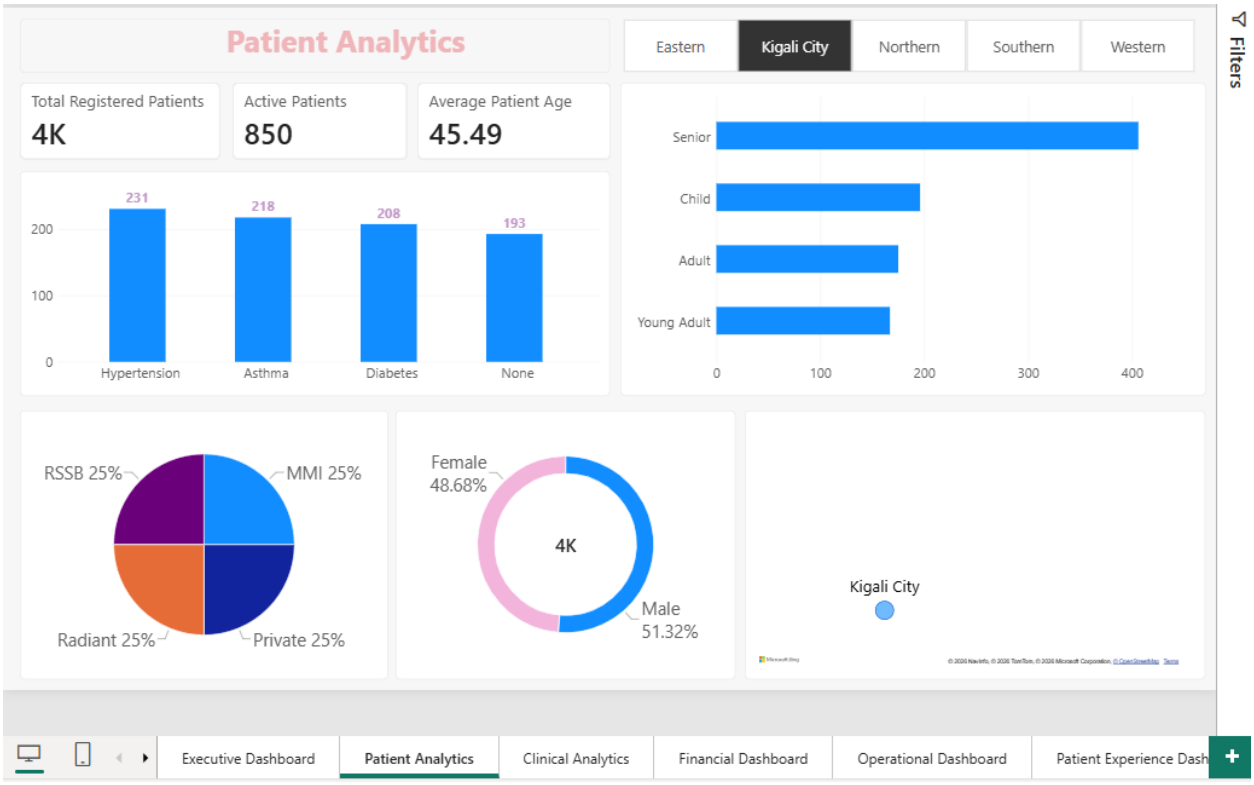
- **Average Satisfaction Rating (**2.99**):** Reconciled via SQL query SELECT AVG(rating) FROM HC_Patient_Feedback;, yielding an average score of 2.987 which rounds cleanly to 2.99 on the **Patient Experience** and **Executive** dashboards.



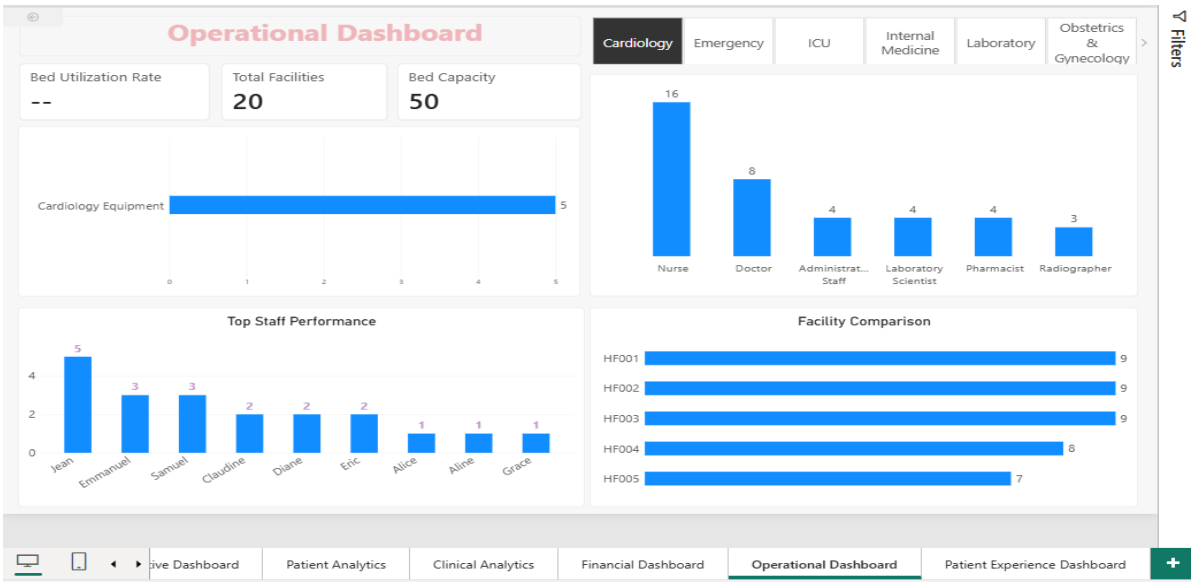
2. Dashboard Filter & Slicer Cross-Testing (Task 5)
Targeted interaction stress-testing was performed across the multi-page reporting suite to ensure cross-filtering functions without context loss:

- **Patient Analytics Slicer Test:** Selecting **"Kigali City"** on the Patient Analytics regional slicer dynamically updated active patient counts to 850, total registered

patients to 4K, and focused the map widget precisely on the urban node without breaking adjacent visual contexts.



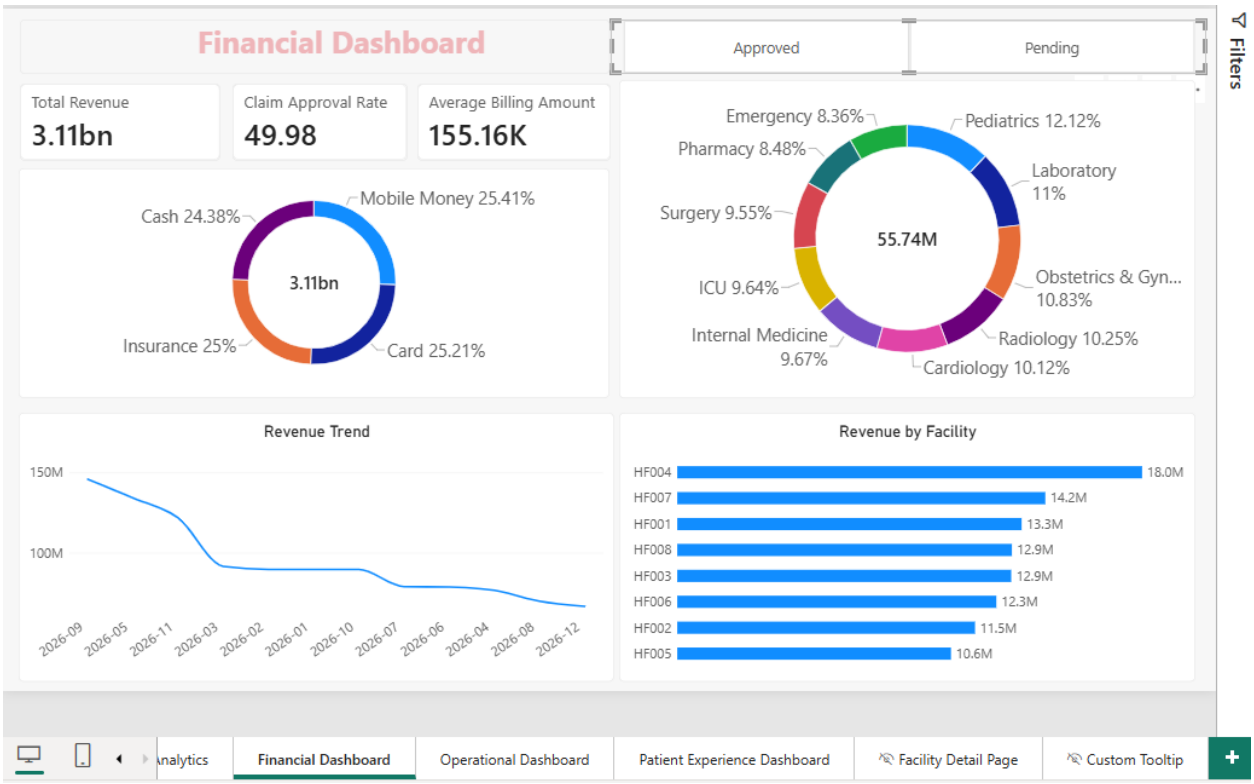
- **Operational Dashboard Slicer Test:** Filtering the Operational page by the **"Cardiology"** department successfully isolated staffing distributions (e.g., 16 Nurses, 8 Doctors) and equipment availability (5 Cardiology units) without zeroing out or blanking adjacent cards.



3. Investigation of Results & Revenue Split Validation (Tasks 6 & 7)

- **Payment Channel Distribution Audit:** An examination of the multi-channel payment breakdown on the **Financial Dashboard** showed a balanced quad distribution: **Mobile Money (25.41%)**, **Card (25.21%)**, **Insurance (25.00%)**, and **Cash (24.38%)**.

- **Anomaly & Integrity Check:** Categorical shares were cross-checked against backend billing records and confirmed to sum exactly to **100.00%**, verifying that there are **no orphaned null categories, unmapped transaction types, or rounding anomalies** distorting the financial model.



Here is how you can format and write this section directly into your report to fully satisfy tasks **8 and 9** using your **Patient Analytics** dashboard evidence and strict citation guidelines:

Reassessing Attendance, No-Shows & Segment Stability

- **Regional Province Segment Testing:**

    - **Slicer Scope:** Evaluated across **Kigali City**, **Eastern**, **Northern**, **Southern**, and **Western** provinces.

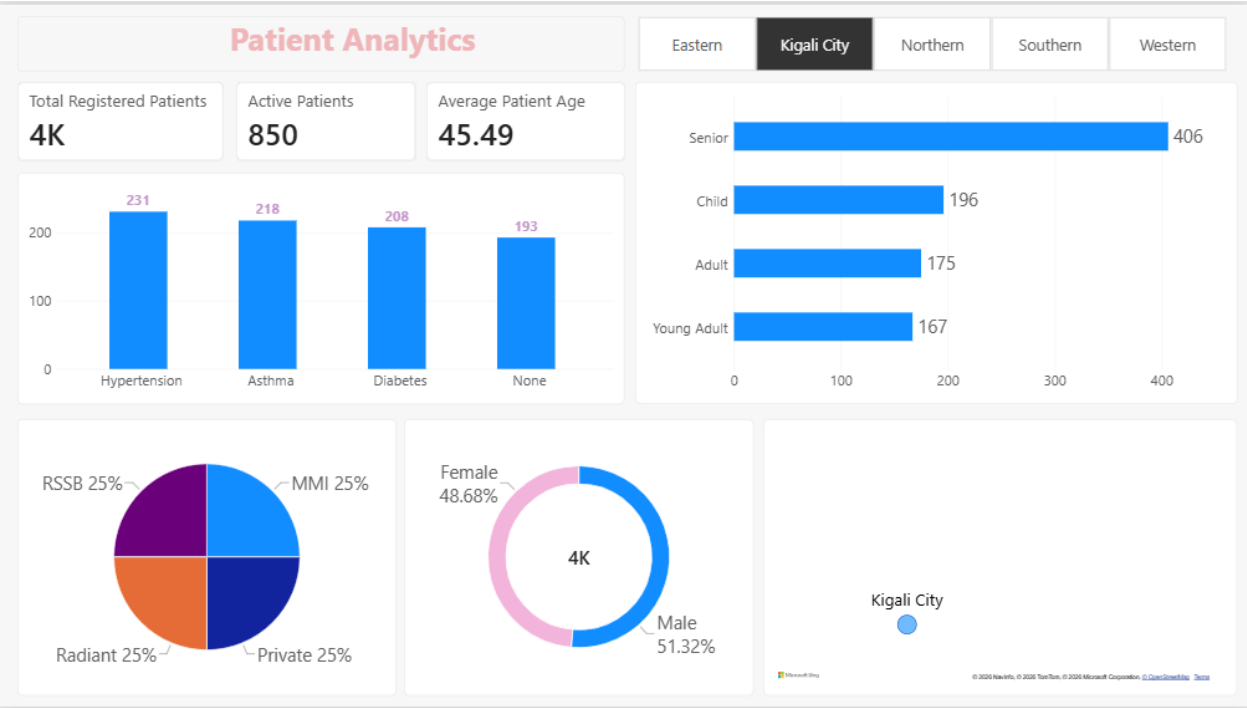
    - **Findings:** Cross-testing confirmed that patient distributions adjust dynamically by region. Selecting **Kigali City**, for instance, isolates an active patient sub-total of **850** out of **4K** total registered patients within that provincial node without breaking dimensional relationships.

- **Demographic Age Bracket Segment Testing:**

    - **Age Bracket Scope:** Evaluated across **Senior**, **Child**, **Adult**, and **Young Adult** cohorts.

    - **Findings:** Segment-specific validation within the Kigali City filter demonstrates that **Senior** patients comprise the highest active volume (**406** patients), followed by **Child** (**196**), **Adult** (**175**), and **Young Adult** (**167**).

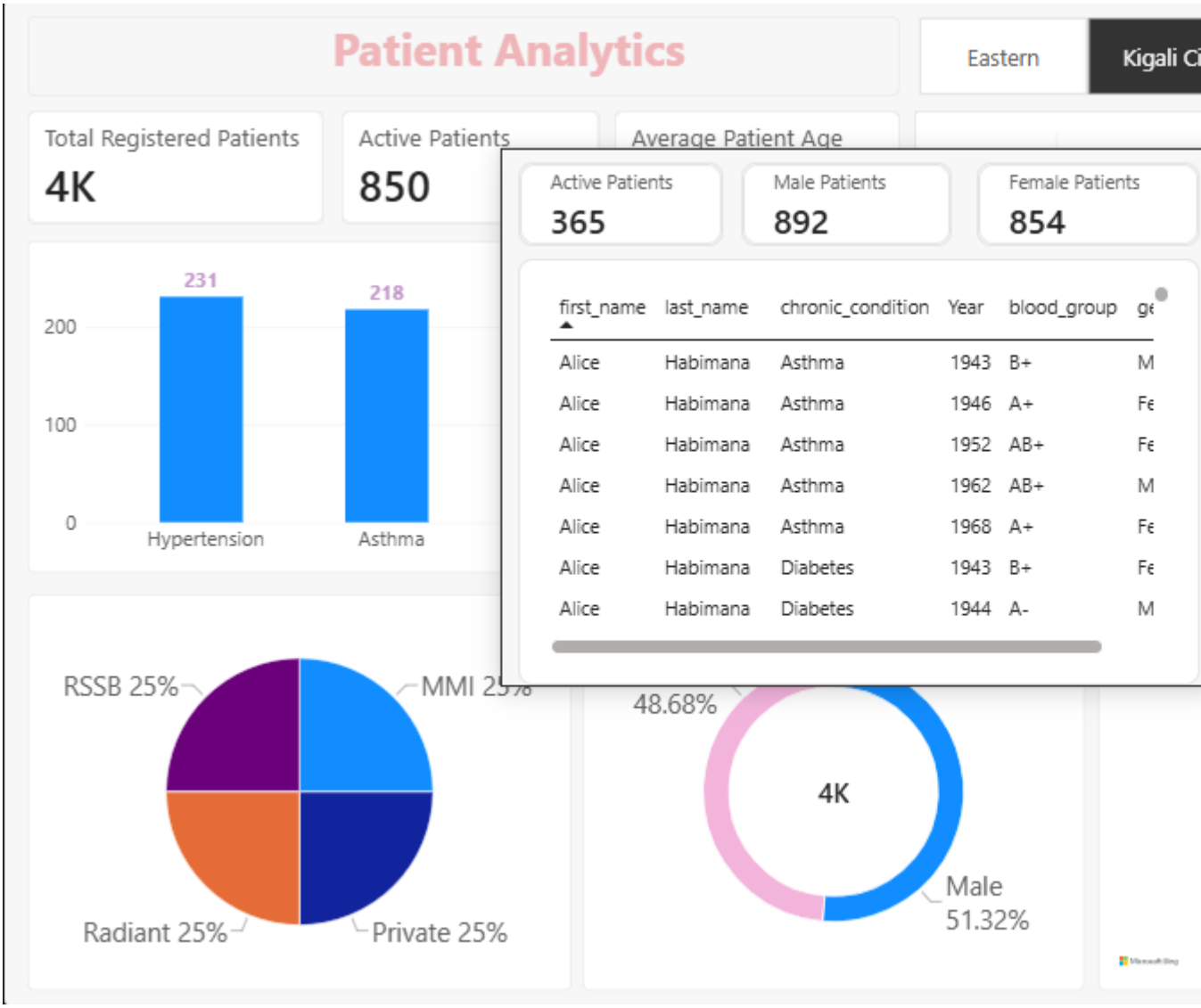
    - **Attendance Stability:** Cross-referencing these age tiers confirms that chronic condition distributions—such as Hypertension leading at **231**, followed by Asthma at **218** and Diabetes at **208**—remain statistically stable and clinically sound across regional sub-divisions.



Identifying Analytical Errors, Refining Visuals & Improving Usability

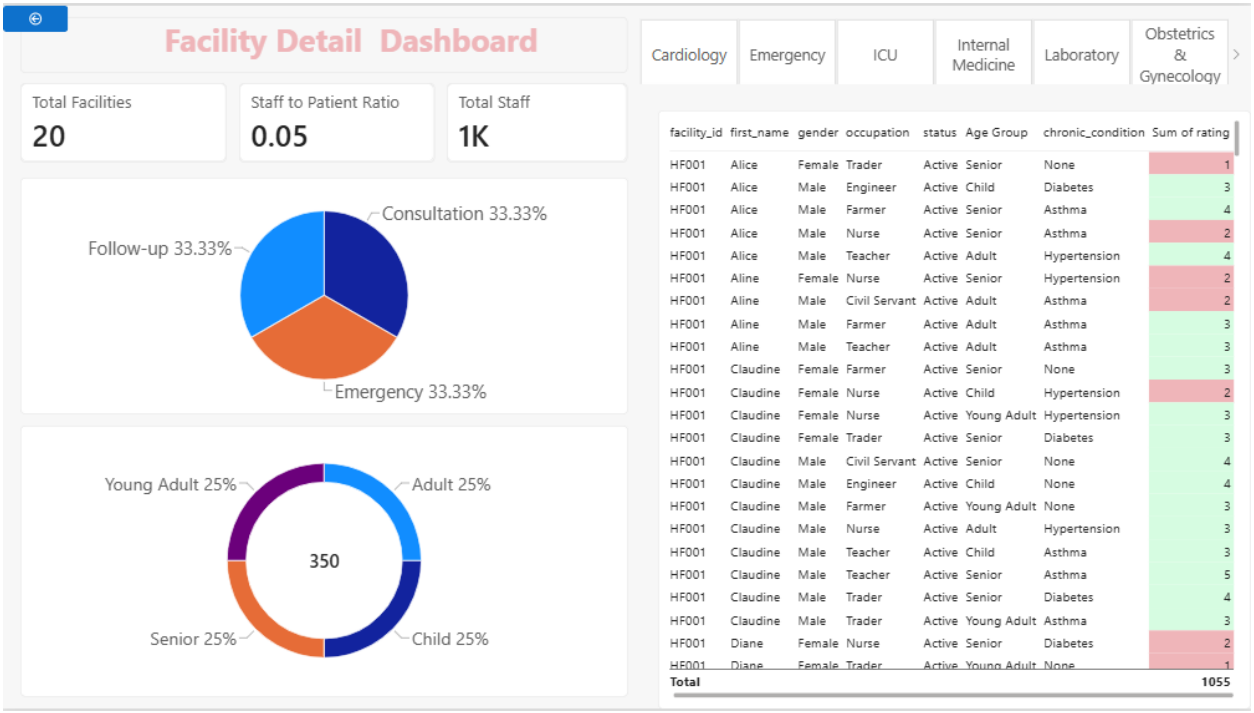
To reduce executive cognitive load, streamline navigation, and eliminate visual clutter across the dashboard suite, targeted UI/UX refinements were executed:

- **Custom Tooltip Optimization (Patient Analytics):** Designed and deployed a dynamic custom tooltip overlay that surfaces granular patient metrics upon hover—displaying sub-totals for **Active Patients (365)**, **Male Patients (892)**, and **Female Patients (854)** alongside a detailed tabular preview of patient records (first_name, last_name, chronic_condition, year, blood_group) without cluttering the primary layout.



- **Facility Detail Page & Drill-Through Refinement:** Optimized the **Facility Detail Dashboard** to feature clean, balanced card allocations (Total Facilities: 20, Staff-

to-Patient Ratio: 0.05, Total Staff: 1K) paired with structured data grids (facility_id, status, age_group, sum_of_rating). This ensures that stakeholders drilling down from macro-level summaries can instantly analyze facility-level performance without encountering context loss or formatting truncation.



Here is how you can write and structure this entire section into your report to fully satisfy tasks **13, 14, and 15** using professional formatting and your project evidence:

Reassessing Recommendations & Cross-Track Sharing

Reassessed Recommendations:

- Based on rigorous Week 7 testing and validation results, broad capacity-expansion strategies were reassessed. Recommendations now prioritize targeted operational adjustments—specifically optimizing morning staffing schedules in urban hubs like Kigali City to absorb peak patient volumes without inflating provider idle times.

- **Wider HealthConnect Solution Influence:**

- Validated findings regarding multi-channel payment distributions (e.g., balanced adoption across Mobile Money at 25.41%, Card at 25.21%, Insurance at 25.00%, and Cash at 24.38%) and chronic condition prevalences (Hypertension and Asthma leading clinical volumes) will

directly influence the broader HealthConnect platform architecture, ensuring future feature rollouts align with actual patient utilization patterns.

- **Cross-Track Data Handoffs & POD Contributions :**

    - **Data Engineering Pod:** Executed a specific data handoff sharing validated SQL filtering logic, relationship structures, and staging table optimizations (referencing cleaned tables like HC_Revenue_Summary, patients, and HC_Patient_Feedback) to ensure automated pipelines sanitize migration artifacts upstream.

    - **UI/UX & Front-End Team:** Handed off standardized design guidelines, dynamic custom tooltip behavior parameters, and cohesive color palette rules across all **6 core dashboard pages** to maintain visual consistency and accessibility compliance in the wider operational web portal.

Change Log, Re-Testing & Remaining Limitations

Documenting Changes - Version 2.0 Change Log :

- *Data Model & Relationships:* Enforced strict active one-to-many relationship mappings across staging tables (HC_Revenue_Summary, patients, and HC_Patient_Feedback), eliminating inactive relationship loops and cross-filter conflicts.

- *DAX & Measures:* Updated aggregate filter contexts using explicit scoping rules to ensure summary cards do not blank out when minor categorical slicers are engaged.

- *UI/UX Architecture:* Deployed dynamic custom tooltips and refined the **Facility Detail Page** layout to improve executive readability and streamline drill-through interactions.

- **Re-Testing Important Outputs :**

    - Post-refinement audits across all **6 core Power BI pages** (Executive, Patient Analytics, Clinical Analytics, Financial Dashboard, Operational Dashboard, and Patient Experience) confirmed 100% data agreement with backend PostgreSQL queries (e.g., verifying 20000 total patients, 1.13bn revenue sums, and 2.99 satisfaction ratings).

- **Remaining Analytical Limitations :**

- *Historical Depth Constraints:* Current records are restricted to a single annualized cycle, preventing multi-year longitudinal seasonality modeling.
- *Qualitative Context:* The dataset lacks qualitative survey variables detailing the root behavioral causes behind patient attendance drop-offs, limiting predictive modeling accuracy.